DO NOW — RECALL

Lesson 055 · Choosing Data Structures for Problems

Name: _____ Class: _____ Date: _____

1. What is a hash collision?

2. When can linked-list insertion be O(1)?

3. What is a BST's advantage when balanced?

4. An empty queue receives 4, 8 and 2 in that order. Repeatedly dequeue an item and push it onto an empty stack until the queue is empty. Trace the sequence returned by popping the stack until it is empty.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Two different keys map to the same bucket.

2. Model answer

When a reference to the insertion node is already known.

3. Model answer

Ordered search can require O(log n) comparisons.

4. Model answer

2, 8, 4. The queue supplies 4, 8, 2; the stack reverses that order.

By the end, you can…

• Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

• Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

• Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

Vocabulary — know these terms

operation — Action such as lookup or insertion.

random access — Direct access by numeric index.

trade-off — A benefit weighed against a cost.

complexity — Growth in resource use.

NEW LEARNING

Read, annotate and explain

Discover

Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

Trace

Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

Evaluate

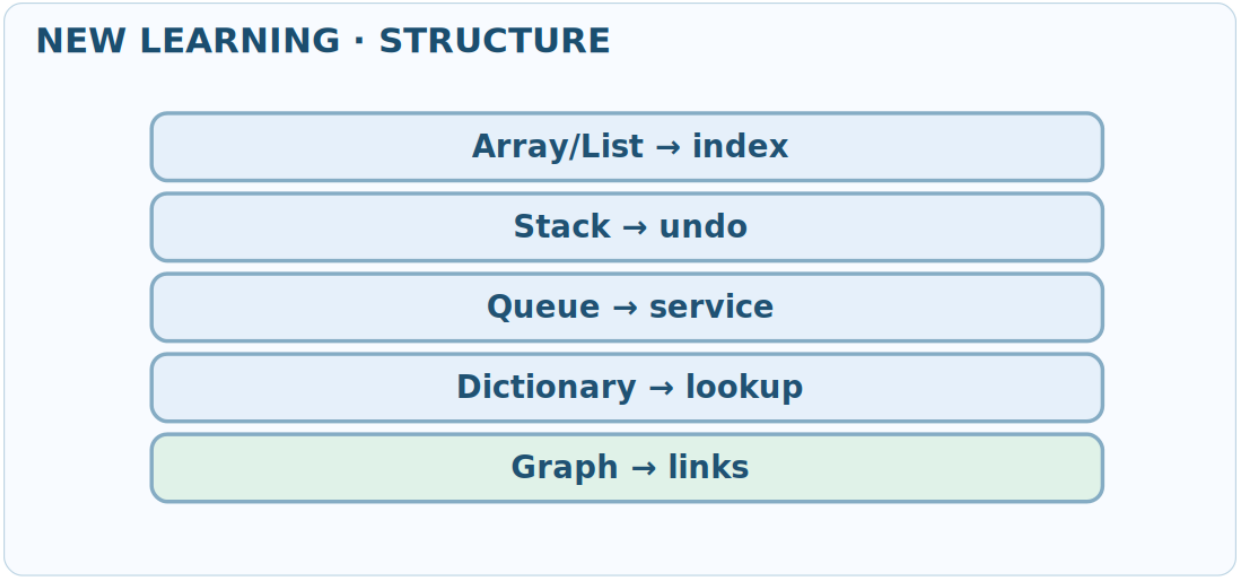
Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · STRUCTURE: Array/List → index, Stack → undo, Queue → service, Dictionary → lookup, Graph → links

I DO — WORKED EXAMPLE

Follow the worked method

Choose a structure for a browser's Back action and justify.

Explain the trace

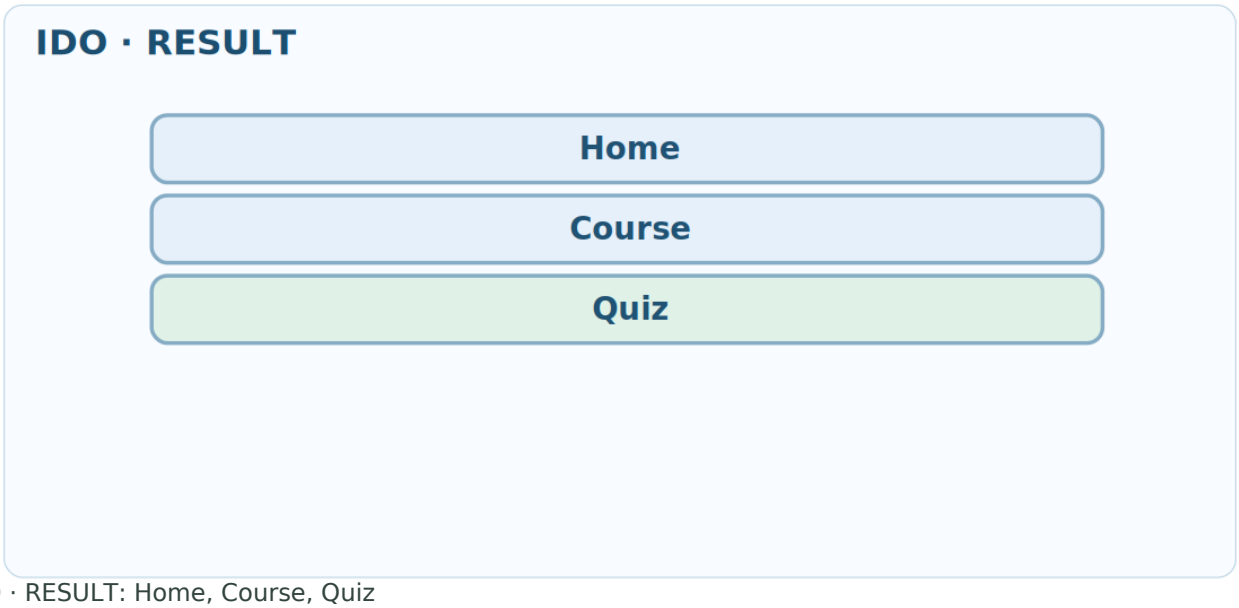
Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose a structure for a browser's Back action and justify.

Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.



IDO · RESULT: Home, Course, Quiz

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Choose a structure for clinic tickets processed in arrival order.
Source code / input:
queue.Enqueue(ticket)
next=queue.Dequeue()

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Match an undo feature, printer queue, pupil ID lookup and road map to a structure. [4]

Question 2

A timetable stores fixed grid of rooms and periods. Recommend a 2D array or graph and explain operations. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate using List<T> versus linked list for a million-item collection, focusing on frequent random index access versus middle insertion with a known node. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

The same data structure is always optimal for every collection.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

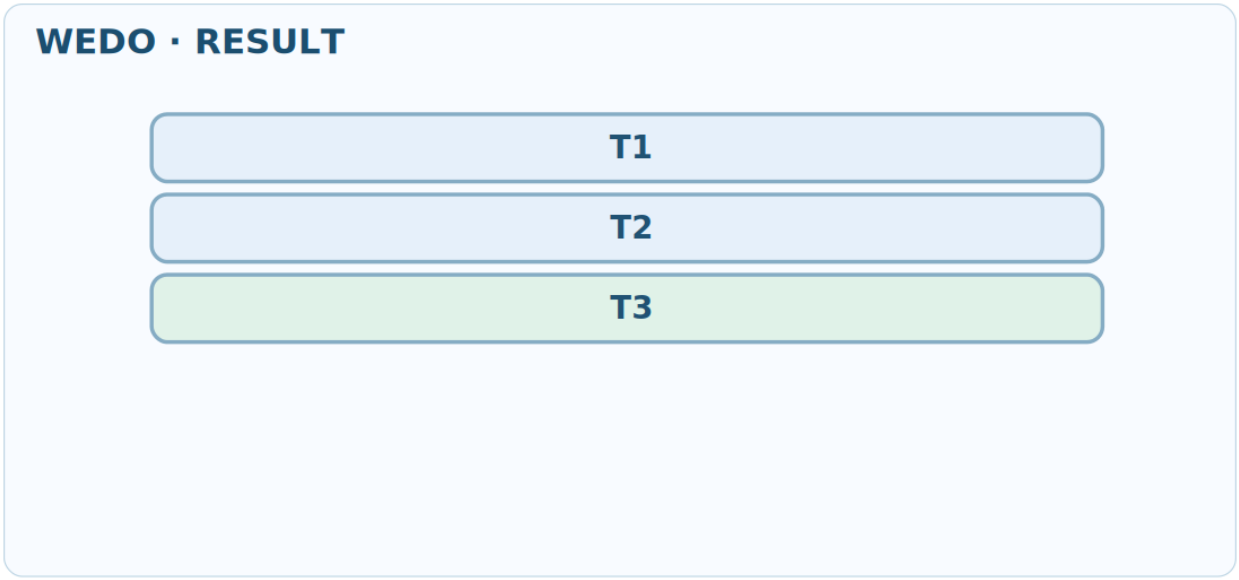
Explain the trace
Queue FIFO serves earliest arrivals first, subject to real-world priority policies.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose a structure for clinic tickets processed in arrival order.

Queue FIFO serves earliest arrivals first, subject to real-world priority policies.



WEDO · RESULT: T1, T2, T3